

Advancements in Machine Learning for Cancer Genomics

Machine learning has become more and more important in cancer genomics as researchers try to interpret multi-dimensional datasets and find patterns and other insights. There are many approaches, however, deep learning specifically has gained popularity due to its ability to learn and recognize complex and nonlinear patterns from this data. This review paper will synthesize two studies that apply this deep learning to cancer genomics: *Assessment of deep learning and transfer learning for cancer prediction based on gene expression data* by Hanczar et al. (2022) and *Deep Learning for Biomarker Discovery in Cancer Genomics* by Wu et al. (2024). Both studies use deep learning, however with different approaches, objectives, data, designs, and biological interpretations. Both papers give a broader perspective on deep learning and its support in cancer classification as well as biomarker discovery.

Background and Study Objectives:

Cancer genomics research relies on machine learning to find patterns regarding tumor type, prognosis, and genetic-based data. Hanczar et al. (2022) looked at how neural networks and learning strategies perform in predicting different cancer types from gene profiles. The goal with their study was to challenge the previous assumptions that deep learning underperforms with “small- n , large- p ” datasets. In this context, n would represent the number of samples (ex. patients) and p represents the number of features (ex. genomic variants). In many cancer studies, p can exceed 20,000-50,000 genes while n might be in the low hundreds or less. This difference makes it easy for machine learning models to overfit, memorizing training samples rather than learning larger recognizable patterns. Hanczar et al. looked to challenge this issue by benchmarking multiple architectures across 36 datasets. They then looked at whether transfer learning (a model trained for one task used as a starting point for another) could increase classification accuracy when training data are limited.

Wu et al. (2024) focused instead on biomarker discovery. They proposed a multiple-instance learning framework designed to find patterns and specifically look for two major genomic instability phenotypes (microsatellite instability and homologous recombination deficiency). The model was trained to do this directly from sequence data. These two genomic features are important predictors of immune checkpoint inhibitor response and provide significant information regarding the immune system as a whole. Their study looked to determine if deep learning could capture mutational signatures without relying on specially designed input features.

While both papers look at applying deep learning to cancer genomic datasets, the goals reflect two different machine learning tasks. Hanczar’s paper focuses on multiclass cancer classification based on expression levels, while Wu’s has more of a focus on phenotype prediction and biomarker discovery based on mutation sequences. This contrast shows how machine learning can have a broad impact within the medical field and can be trained to help in many disciplines. It also demonstrates how variation within data, objective, and clinical relevance can shape a model completely differently.

Data Types and Model Design:

The two studies differ the most in their choice of genetic data. Hanczar et al. analyzed gene expression matrices, which are large datasets where each sample corresponds to tens of thousands of transcript measurements. These datasets are high-dimensional but typically have

very few samples. This means that authors must evaluate convolutional neural networks (models that specialize in looking at analyzing visual data such as images) and transfer learning practices that use knowledge from previously related gene expression tasks.

Wu et al. used mutation sequence data from variable-length lists of nucleotide changes across the genome. Normal deep learning models cannot process sample-specific sets of mutations directly, so the authors designed a model where each genetic mutation was treated as an “instance” belonging to a “bag” representing a patient’s tumor. This design allowed the model to learn which subsets of mutations contributed most strongly to the two major genomic instability phenotypes, supporting biomarker discovery.

Hanczar et al. evaluated many model architectures and concluded that simpler neural networks often match or do better than the performance of deeper models because cancer expression datasets are often very small with few samples. Transfer learning improves performance only when source and target datasets share biological similarities. The findings from this research reinforced the known limitations of deep learning in low-sample settings, but also proved that careful model setup with a specific objective and training choices can result in successful expression-based predictors.

Wu et al.’s model was a bit more complex but also necessary based on the nature of the mutation data used in their research. Since genomic instability phenotypes come from specific mutational patterns spread across the genome, an attention-based approach was necessary for capturing sparse but crucial signals. This approach allows models to focus on the most relevant parts of input data by assigning weights.

Results and Limitations:

Hanczar et al. reported that simpler neural network models frequently outperform deeper and more complex models for cancer classification using gene expression data. These simpler models often generalize data better because they have fewer parameters and reduce the risk of overfitting in the “small n, large p” setting. Transfer learning improves the classification performance in some cases but not all, because gene expression datasets can vary widely in platform, normalization, and other biological context. A strength of their study was the benchmarking across many independent datasets, which provided a realistic assessment of generalizability. Their work is limited by the lack of interpretability to traditional neural networks, as well as the challenges that come with transferring models across different datasets.

Wu et al. found that their multiple-instance learning model achieved higher predictive accuracy for microsatellite instability and homologous recombination deficiency than baseline machine learning methods and other traditional approaches. One of the strengths of their work was the interpretability, which helped to identify genomic patterns easily. Another strength of this model in the study was that it does not require preemptive information, making it very adaptable to other sequencing. The approach does require high-quality sequencing to perform well, and the mutation patterns can be influenced by technical variation and laboratory bias.

Future Directions:

These studies show the versatility as well as the challenges that come with deep learning models in cancer genomics. Deep learning can both classify cancers from gene expression, and more complex sequence-based models can identify biomarker patterns relevant to clinical work. The contrast between these two papers shows that model selection is crucial when looking at the structure of the data and the biological question of the research. Future research should focus on

integrating expression, mutation, and other data into deep learning frameworks. Improved and larger training datasets would be essential to translate deep learning into clinical applications.

Works Cited:

Hanczar, B., Dunbar, J., & Nadif, M. (2022). Assessment of deep learning and transfer learning for cancer prediction based on gene expression data. *BMC Bioinformatics*, 23, 1–19.

<https://doi.org/10.1186/s12859-022-04807-7>

Wu, S., Liu, Y., Zeng, T., & Jiang, R. (2024). Deep learning for biomarker discovery in cancer genomes. *Bioinformatics*, 40(2), btae050. <https://pubmed.ncbi.nlm.nih.gov/39829845/>